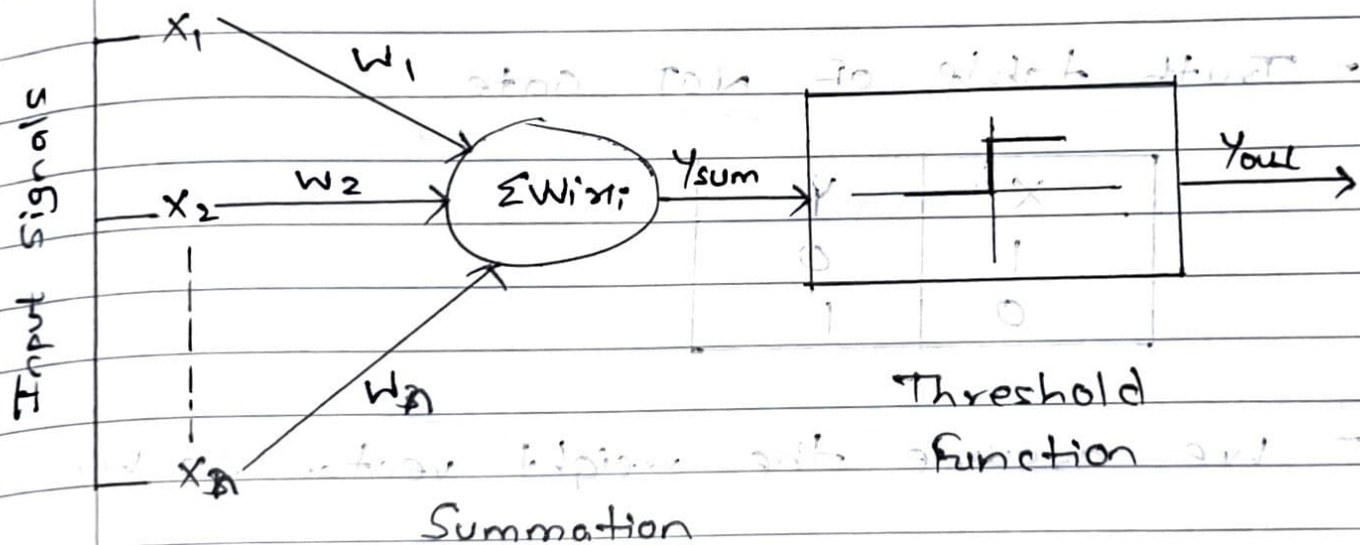


Q.1 Explain Mc-Culloch Pitts Model.

⇒

- The McCulloch Pitts neural model, which was the earliest ANN model, has only two types of inputs — Excitatory and Inhibitory.
- The excitatory inputs have weights of positive magnitude and inhibitory weights have weights of negative magnitude.
- The inputs of the McCulloch-Pitts neuron could be either 0 or 1.
- It has a threshold function as an activation function.



- So, the output signal  $y_{out}$  is 1 if the input  $y_{sum}$  is greater than or equal to a given threshold value, else 0.
- The diagrammatic representation of the model is shown above.

- The output of above model can be written as,

$$Y = \begin{cases} 1 & \text{if } \sum W_i x_i \geq \theta \\ 0 & \text{if } \sum W_i x_i < \theta \end{cases}$$

- Simple Mc-Culloch Pitts neurons can be used to design logical operations.
- For that purpose, the connection weights need to be correctly decided along with the threshold function.
- For e.g.,  
Simulation of NOT gate.

- Truth table of NOT gate

| X | Y |
|---|---|
| 1 | 0 |
| 0 | 1 |

- We assume the weight vector as  $W$ .
- For the first row (i.e.  $X=1$ ) we may write net value as  $W * X = W * 1 = W$  according to the model if the output is 0 then net value must be less than 0 threshold.  $W < 0$ .
- For the second row (i.e.  $X=0$ ) we may write net value as  $W * X = W * 0 = 0$  according to the model if the output is 1 then net value must be greater than or

equal to threshold,  $\underline{0 \geq \theta}$

- Now we are having two equations

1.  $W < \theta$

2.  $0 \geq \theta$

- Now select the value of  $W$  and  $\theta$  such that above conditions gets satisfied.

- One of the possible values are  $\theta = 0$ ,  $W = -1$ .

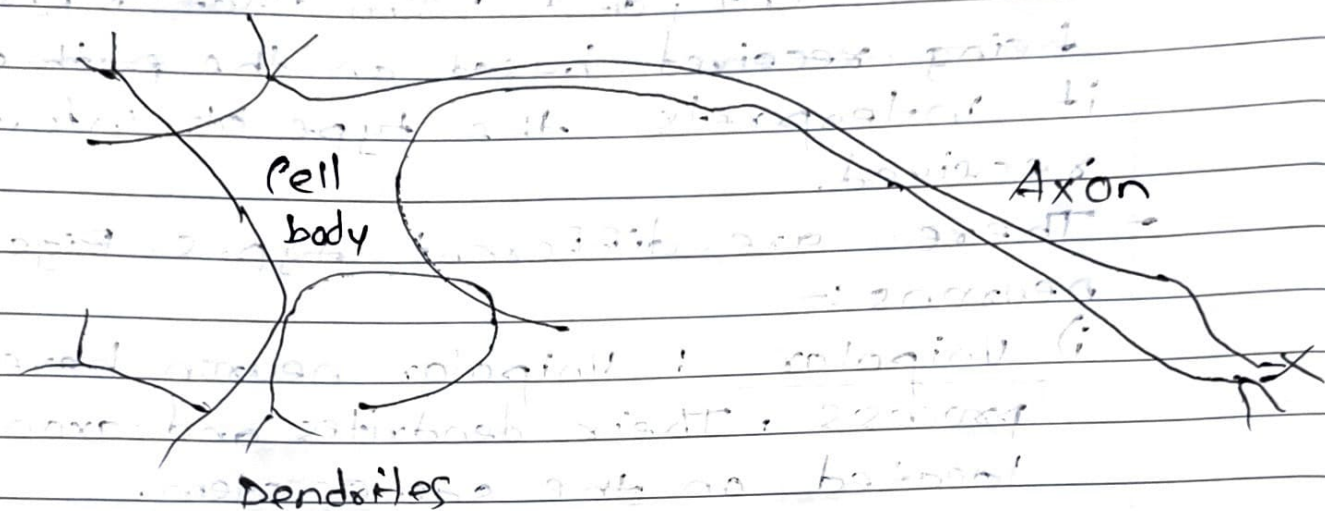
- Using this value NOT gate is represented as,



Q.2 Explain biological neuron. Difference between biological and artificial neuron.

⇒

- Biological Neural Network (BNN) is a structure that consists of synapse, dendrites, cell body and axon.
- In this neural network, the processing is carried out by neurons.
- Dendrites receive signals from other neurons, soma sums all the incoming signals and axon transmits the signal to other cell.



- Dendrites accept the input from the neighbouring neuron through axon.
- Dendrites look like a tree structure, receives signals from other neurons.
- Synapse is the electro-chemical contact between the organs. They do not physically touch because they are separated by a cleft.
- The electric signals are sent through the chemical interactions.

## ANN

1) Structure :-  
 input  
 weight  
 output  
 hidden layers.

2) Information is processed at a faster rate.

3) The response time is measured in nanoseconds.

4) Learning is very precise structures and formatted data.

5) Signal type is numeric value.

## BNN

1) Structure :-  
 dendrites  
 synapse  
 axon  
 cell body

2) Processes information at a slower rate.

3) The response time is measured in milliseconds.

4) They can tolerate ambiguity.

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axon

cell body

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### Q.3 Simulation of AND gate using McCulloch-Pitts model.

⇒

- The truth table of AND gate is as follows :-

| Input |       | Output |
|-------|-------|--------|
| $x_1$ | $x_2$ | $y$    |
| 0     | 0     | 0      |
| 0     | 1     | 0      |
| 1     | 0     | 0      |
| 1     | 1     | 1      |

- We assume the weight vector  $w_1$  for  $x_1$  and  $w_2$  for  $x_2$ .

- For 1<sup>st</sup> row,

$$\begin{aligned} \text{net} &= (w_1 * x_1) + (w_2 * x_2) \\ &= 0 \end{aligned}$$

According to MP model, if the output is 0 then net value must be less than threshold,  $0 < \theta$

- For 2<sup>nd</sup> row,

$$\begin{aligned} \text{net} &= (w_1 * x_1) + (w_2 * x_2) \\ &= w_2 \end{aligned}$$

According to MP model,  
 $w_2 < \theta$

- For 3<sup>rd</sup> row,

$$\text{net} = (W_1 * x_1) + (W_2 * x_2)$$

$$\Rightarrow W_1$$

According to MP model,

$$W_1 < \theta$$

- For 4<sup>th</sup> row,

$$\text{net} = (W_1 * x_1) + (W_2 * x_2)$$

$$= W_1 + W_2$$

According to MP model,

$$W_1 + W_2 \geq \theta$$

- Now we have four conditions

$$1) \quad 0 < \theta$$

$$3) \quad W_1 < \theta$$

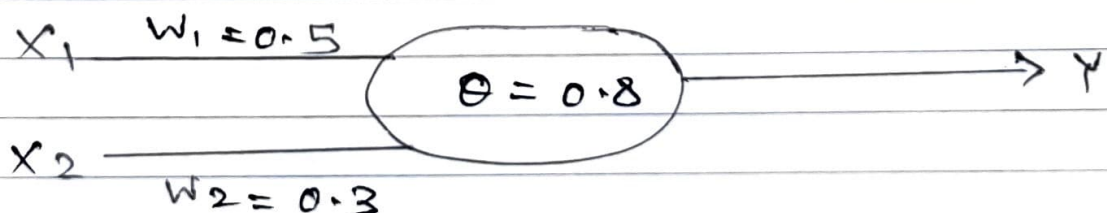
$$2) \quad W_2 < \theta$$

$$4) \quad W_1 + W_2 \geq \theta$$

- Now select the values of  $W_1$ ,  $W_2$  and  $\theta$  such that above condition get satisfied.

- Let  $\theta = 0.8$ ,  $W_2 = 0.3$ ,  $W_1 = 0.5$

- AND Gate is represented as,



Q.4

Design NAND gate using Mc-culloch Pitts model.

⇒

The truth table of NAND gate is as follows :-

| Input |       | Output |
|-------|-------|--------|
| $X_1$ | $X_2$ | $Y$    |
| 0     | 0     | 1      |
| 0     | 1     | 1      |
| 1     | 0     | 1      |
| 1     | 1     | 0      |

We assume weight vector is  $W_1$  for  $X_1$  and  $W_2$  for  $X_2$ .

The output of MP model is given as,

$$F(y_{in}) = \begin{cases} 1 & \text{if } y_i \geq \theta \\ 0 & \text{if } y_i < \theta \end{cases}$$

For the first row,

$$(W_1 \times X_1) + (W_2 \times X_2) = 0$$

According to the MP model if output is 1, then net value must be greater than equal to  $\theta$ .

$$0 \geq \theta$$

- For the second row,

$$(W_1 * x_2) + (W_2 * x_2) = 0 + W_2 = W_2$$

According to the MP model,

$$W_2 \geq 0$$

- For the third row,

$$(W_1 * x_1) + (W_2 * x_2) = W_1 + 0 = W_1$$

According to MP model,

$$W_1 \geq 0$$

- For the fourth row,

$$(W_1 * x_1) + (W_2 * x_2) = W_1 + W_2$$

According to MP model, if output is 0 then net value must be less than threshold

$$W_1 + W_2 < 0$$

- Let's assume  $\theta = -3$ ,  $W_1 = -2$ ,

$$W_2 = -2$$

- We have four conditions :-

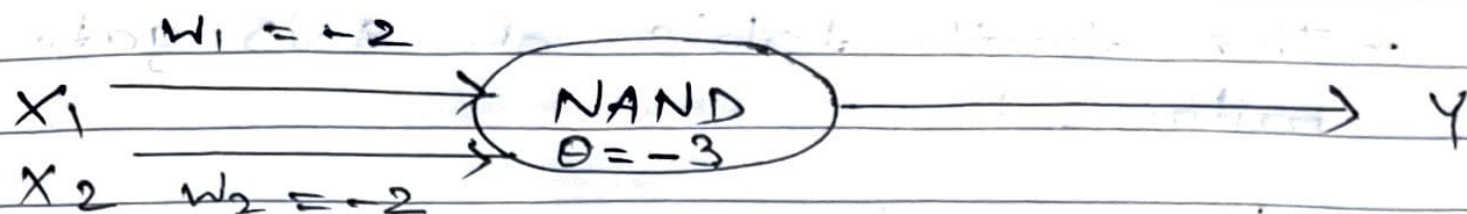
$$1) \quad 0 \geq \theta$$

$$2) \quad W_2 \geq \theta$$

$$3) \quad W_1 \geq \theta$$

$$4) \quad W_1 + W_2 < \theta$$

- Using this value NAND gate is represented as,



Q.5 Explain McCulloch Pitts Model. Design simulation of NOR gate & NAND gate using McCulloch Pitts Model.

⇒

The truth table of NOR gate is as follows:-

| Input |       | Output |
|-------|-------|--------|
| $x_1$ | $x_2$ | $y$    |
| 0     | 0     | 1      |
| 0     | 1     | 0      |
| 1     | 0     | 0      |
| 1     | 1     | 0      |

- We assume weight vector as  $w_1$  for  $x_1$  and  $w_2$  for  $x_2$ .

- For 1<sup>st</sup> row,

$$\begin{aligned}\text{net} &= (w_1 * x_1) + (w_2 * x_2) \\ &= 0\end{aligned}$$

According to MP model, if the output is 1 then net value must be greater than equal to  $\theta$ .

$$\text{i.e. } 0 \geq \theta$$

- For 2<sup>nd</sup> row,

$$\begin{aligned}\text{net} &= (w_1 * 0) + (w_2 * 1) \\ &= w_2\end{aligned}$$

- According MP model, IF output is 0 the net value must be less than 0.  
i.e.

$$W_2 < 0$$

- For third row,

$$\text{net} = (W_1 \times 1) + (W_2 \times 0)$$

$$= W_1$$

i.e.  $W_1 < 0$

- For fourth row,

$$\text{net} = (W_1 \times 1) + (W_2 \times 1)$$

$$= W_1 + W_2$$

i.e.  $W_1 + W_2 < 0$

- We have four conditions

i)  $0 > 0$

ii)  $W_2 < 0$

iii)  $W_1 < 0$

iv)  $W_1 + W_2 < 0$

- Let's assume,

$$\theta = -0.5$$

$$W_1 = W_2 = -1$$

- NOR gate is represented as -

$$X_1 \quad W_1 = -1$$

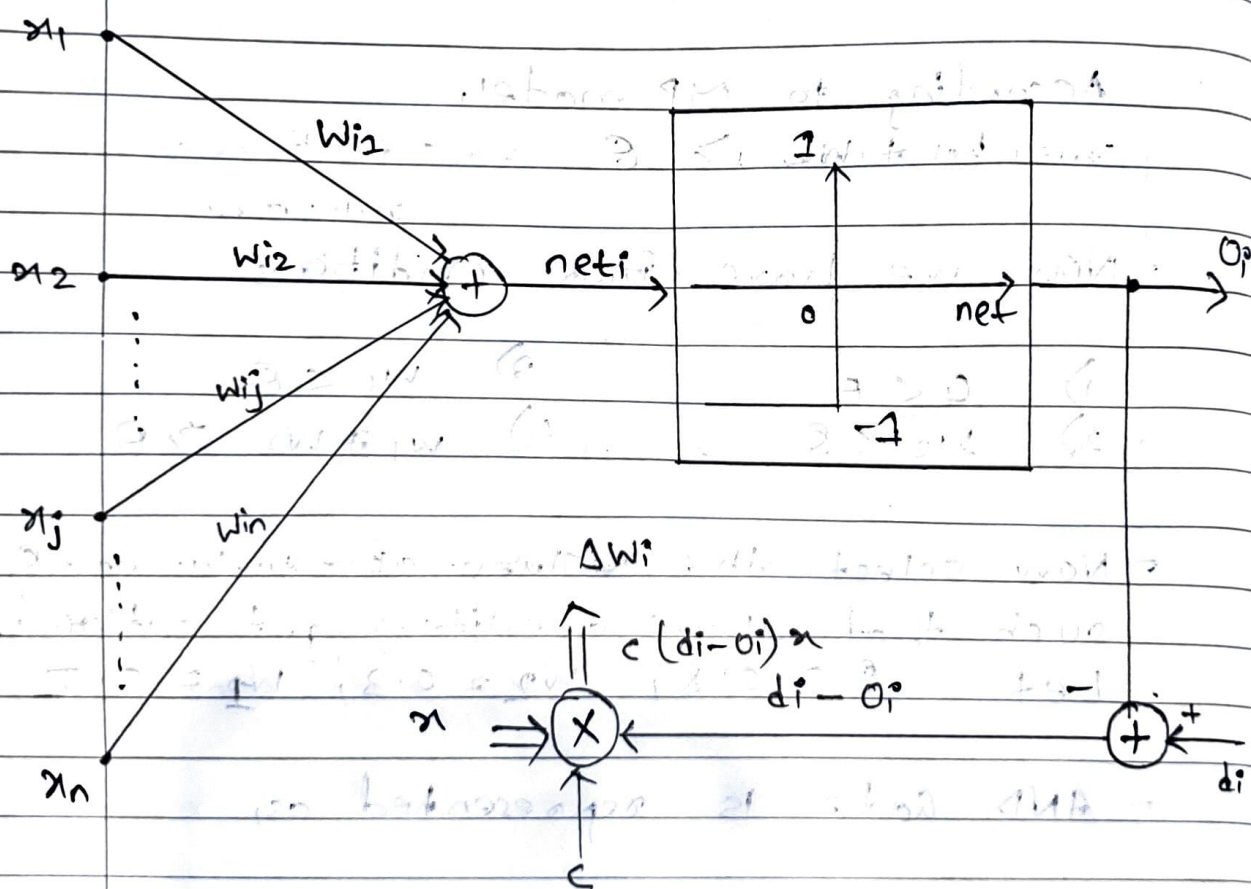
$$X_2 \quad W_2 = -1$$

$$\theta = -0.5$$

$\rightarrow Y$

Q.5 Explain discrete perceptron learning algorithm with block diagram.

- ⇒
- The perceptron learning algorithm is a supervised learning algorithm for binary classification, typically used in neural networks to train a single-layer perceptron.
  - The perceptron aims to find the linear optimal decision boundary that separates two classes.



- Working:

Given are  $P$  training pairs  
 $\{x_1, d_1, x_2, d_2, \dots, x_P, d_P\}$  where

$$x_i = (n \times 1)$$

$$d_i = (1 \times 1), \quad i = 1, 2, \dots, P$$

Step 1:  $c > 0$  is chosen

Step 2:

Weights are initialized at  $w$  at small random values,  $w$  is  $(n+1) \times 1$

Counters and errors are initialized.

$$k \leftarrow 1, \quad p \leftarrow 1, \quad E \leftarrow 0$$

Step 3: The training cycle begins here.

Input is presented and output is computed -

$$y \leftarrow y_p, \quad d \leftarrow d_p$$

$$o \leftarrow \text{sgn}(w^t \cdot y)$$

Step 4: Weights are updated

$$W \leftarrow W + \frac{1}{2} c (d - o) y$$

Step 5: Cycle error is computed

$$E \leftarrow \frac{1}{2} (d - o)^2 + E$$

Step 6: IF  $p < P$ , then  $p \leftarrow p + 1$ ,  
 $k \leftarrow k + 1$  and goto step 3  
 otherwise goto step 7.

Step 7: The training cycle is completed.

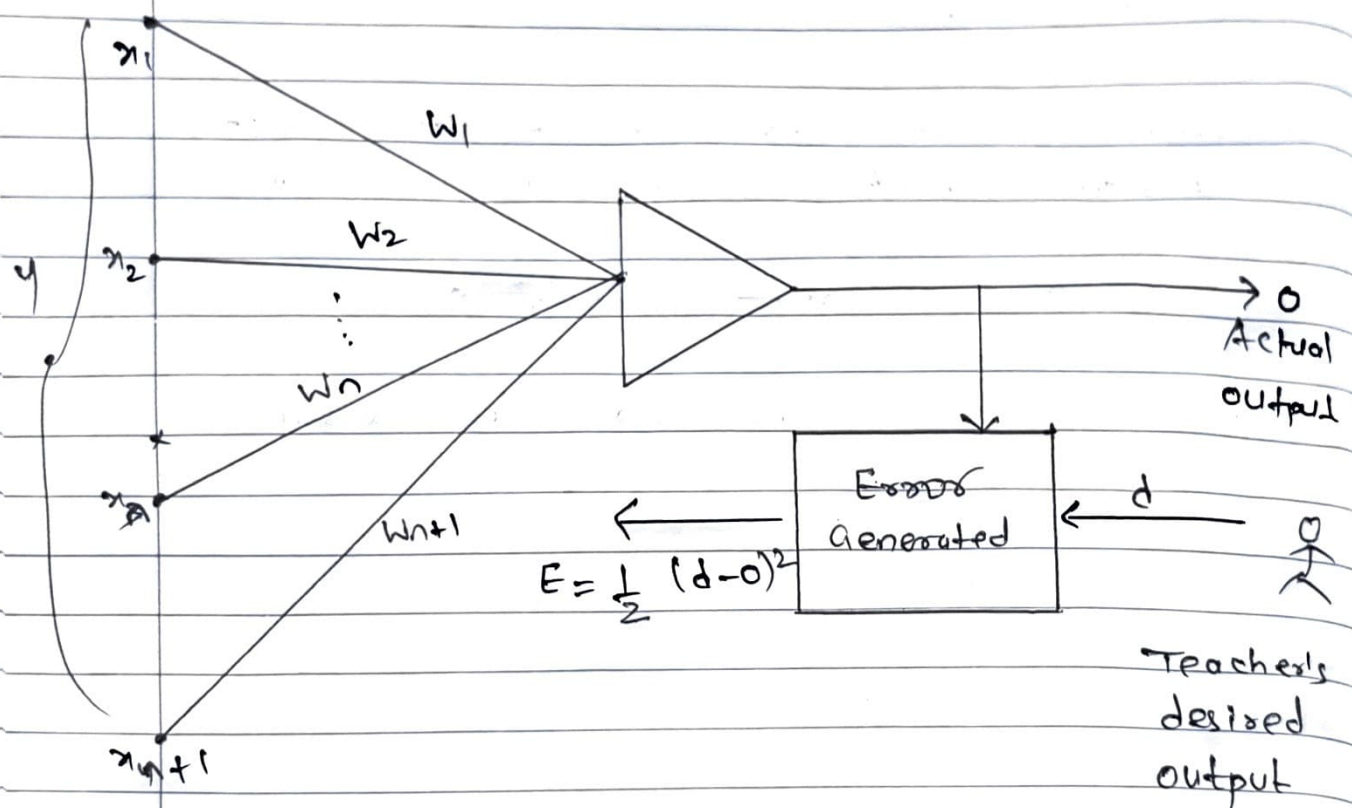
For  $E = 0$ , terminate the training session.

IF  $E > 0$ , then  $E \leftarrow 0$ ,  $p \leftarrow 1$  and enter the new training cycle by going to step 3.

Q.4 Explain continuous perceptron learning algorithm with block diagram.

⇒

- The perceptron learning algorithm is a supervised learning algorithm used for binary classification.
- In its continuous form, the perceptron updates weights using a continuous activation function rather than a step function.
- Block diagram:



- Algorithm:

Given are  $P$  training pairs,

$$\{x_1, d_1, x_2, d_2, \dots, x_p, d_p\}$$

Where,

$$x_i = (n \times 1)$$

$$d_i = (1 \times 1), \quad i = 1, 2, \dots, p$$

Step 1:  $\lambda > 0, \lambda = 1, E_{max} = 0$

Step 2: Weights are initialized at  $w$  at small random values,  $w$  is  $(n+1) \times 1$ .

Step 3: The training cycle begins here. Input  $y$  is presented and output computed:

$$y \leftarrow y_p$$

$$d \leftarrow d_p$$

$$o \leftarrow F(w^t \cdot y)$$

Step 4: Weights are updated.

$$E \leftarrow \frac{1}{2} (d - o)^2 + E$$

$$W \leftarrow W + \frac{1}{2} n (d - o) (1 - o^2) y$$

Step 5: Cycle error is computed.

$$E \leftarrow \frac{1}{2} (d - o)^2 + E$$

Step 6: IF  $p < P$ , then  $p \leftarrow p + 1$ ,  
 $k \leftarrow k + 1$  and goto step 3 otherwise  
 step 7.

Step 7: The training cycle is completed.

For  $E < E_{max}$  terminate the training  
 session. Output weights,  $k$  and  $E$ .

IF  $E \geq E_{max}$ , then  $E \leftarrow 0$ ,  $p \leftarrow 1$   
 and enter the new training cycle by  
 going to step 3.

$$\begin{aligned} p &\rightarrow p \\ k &\rightarrow k \end{aligned}$$

$$(p, tw) \rightarrow 0$$

Step 4: Compute the error

$$E \leftarrow \frac{1}{2} (d - o)^2 + E$$

Q.8 What are the Three classes of Deep Learning, explain each?

- ⇒
- Deep learning refers to a rather wide class of machine learning techniques and architectures, with the hallmark of using many layers of non-linear information processing stages that are hierarchical in nature.
  - Three primary classes are :-
    - 1) Deep network for unsupervised learning
    - 2) Deep network for supervised learning
    - 3) Hybrid Deep Networks

### 1) Deep Network for Unsupervised Learning

⇒ These models are intended to capture high-order correlation of observed data for pattern analysis when no information about target class label is available.

- Examples of such deep networks are Deep autoencoders, Deep Belief Networks (DBNs), Generative Adversarial Network (GANs).

a) Deep Autoencoder : A Deep Neural Network whose output target is the data input itself, often pre-trained with DBN or using distorted training data to regularize the learning.

b) Deep Belief Network : Probabilistic generative models composed of multiple layers of stochastic hidden variables.

c) GAN: A GAN is a deep neural network framework which is able to learn from a set of training data and generate new data with the same characteristics as the training data.

2) Deep Network for supervised learning

⇒ These are intended for exploiting power of discrimination for pattern classification purposes.

- Target label data are always available in direct or indirect forms for such supervised learning.

- They are also called as discriminative deep networks.

- Examples of deep networks that uses supervised learning are CNN, RNN and LSTM.

3) Hybrid Deep Networks

⇒ The term hybrid refers to the deep architectures that either comprises or makes use of both generative and discriminative model components.

- Usually a generative unsupervised deep network is used to help with discrimination, which is the final goal of the hybrid architecture.

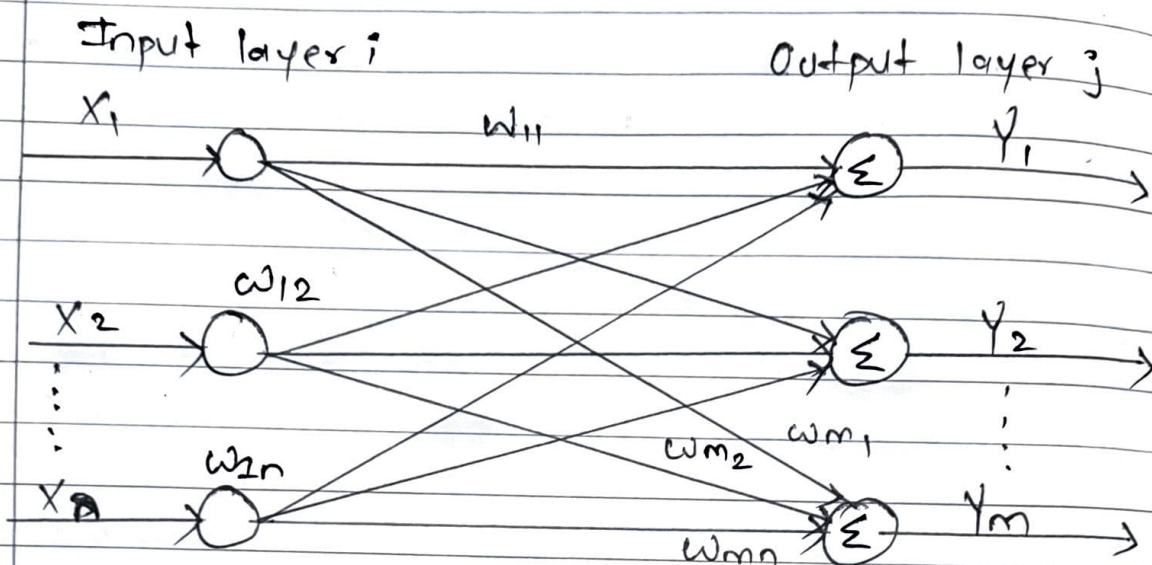
- For example the generative models can be trained in an unsupervised fashion to provide excellent initial parameters for discriminative model.

- A further example of hybrid deep networks is the use of generative models to pre-train deep convolutional neural networks.

Q.9 What are Feed neural networks?

⇒

- In this type of network the neurons are connected in a forward direction.
- Connection is allowed from layer  $i$  to layer  $j$ , if and only if,  $i < j$ .



- Input vector, output vector and the weight matrix of the network are represented as follows,

Input vector,  $X = [x_1, x_2, x_3, \dots, x_n]$   
 Output vector,  $Y = [y_1, y_2, y_3, \dots, y_m]$

Weight matrix,  $W =$

|          |          |          |         |          |
|----------|----------|----------|---------|----------|
| $w_{11}$ | $w_{12}$ | $w_{13}$ | $\dots$ | $w_{1n}$ |
| $w_{21}$ | $w_{22}$ | $w_{23}$ | $\dots$ | $w_{2n}$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\dots$ | $\vdots$ |
| $w_{m1}$ | $w_{m2}$ | $w_{m3}$ | $\dots$ | $w_{mn}$ |

- The <sup>net</sup> input to the  $j^{\text{th}}$  is calculated using the following equation,

$$\text{net}_j = \sum W_{ji} * x_i \quad \text{For } j=1 \text{ to } m \text{ and } i=1 \text{ to } n$$

- Once the net input is calculated, the output of the  $j^{\text{th}}$  neuron is calculated by applying the activation function on the net input as follows,

$$y_j = f(\text{net}_j)$$

### Types:

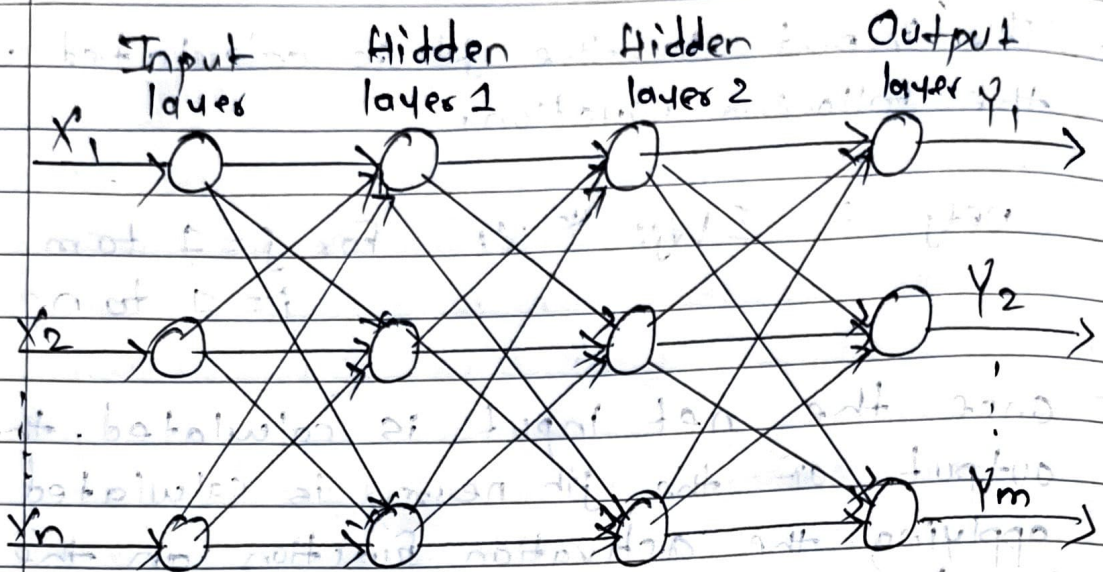
#### 1) Single Layer Perceptron Network

⇒ Above figure represents the single layer perceptron network.

- In this type of network there are only input and output layers are present.
- It consists of single layer, where the inputs are directly connected to the outputs via a series of weights.

#### 2) Multi-layer Perceptron Network

⇒ This type of network besides having the input and output layers also have one or more hidden layers in between input and output layers.



(1) Input layer

(2) Hidden layer

(3) Output layer

The diagram shows a feedforward neural network with four layers: Input layer, Hidden layer 1, Hidden layer 2, and Output layer. The input layer has  $n$  nodes, Hidden layer 1 has 3 nodes, Hidden layer 2 has 3 nodes, and the output layer has  $m$  nodes. All nodes in adjacent layers are fully connected.

The diagram shows a feedforward neural network with four layers: Input layer, Hidden layer 1, Hidden layer 2, and Output layer. The input layer has  $n$  nodes, Hidden layer 1 has 3 nodes, Hidden layer 2 has 3 nodes, and the output layer has  $m$  nodes. All nodes in adjacent layers are fully connected.

Q.10 State any five applications of DL.

1) Automatic Text Generation

⇒ Corpus of text is learned and from this model new text is generated, word-by-word or character-by-character.

- Then this model is capable of learning how to spell, punctuate, form sentences, or it may even capture the style.

2) Image Recognition

⇒ Recognizes and identifies peoples and objects in images as well as to understand content and context.

- This area is already used in Gaming, Retail, Tourism, etc.
- Used in facial recognition for security and attendance system.

3) Recommendation systems

⇒ Suggests products, movies or music based on user behavior.

- Analyzes user preferences and patterns using collaborative filtering and deep neural network.
- Helps increase user engagement and satisfaction through personalized recommendations.

#### 4) Healthcare

- ⇒ Detects diseases like cancer, pneumonia and diabetic retinopathy using DL on medical images.
- Assists doctors in automated report generation and diagnosis.
- Enables personalized treatment plans using patient data analysis.

#### 5) Autonomous Vehicles

- ⇒ Processes real-time data from cameras and sensors for decision-making.
- Performs lane detection and obstacle avoidance using CNNs.
- Assists in path planning and navigation in self-driving cars.

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Q. 4 Problems based on discrete/continuous perceptron learning algorithm.

$$i) \quad X_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix}, \quad X_3 = \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix}$$

$$W_i = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0.5 \end{bmatrix}$$

$$c = 0.1$$

$$d_1 = -1$$

$$d_2 = -1$$

$$d_3 = 1$$

Find weight vector using perceptron learning rule.

⇒

Sol<sup>n</sup>:-

For perceptron learning rule,

$$net_i = W_i^T X$$

$$O_i = \text{sign}(net_i)$$

$$\Delta W_i = c(d - O_i) X$$

Step 1 : Take first training pair  
 $X = X_1, \quad d = d_1$

$$net_1 = W_i^T X$$

$$= \begin{bmatrix} 1 & -1 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix} = 2.5$$

$$O_1 = \text{sign}(\text{net}_1)$$

$$= \text{sign}(2.5)$$

$$O_1 = 1$$

$$\Delta W_1 = c(d_1 - O_1) \times$$

$$= 0.1(-1 - 1)$$

$$\begin{bmatrix} 1 \\ -2 \\ 0 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.2 \\ 0.4 \\ 0 \\ 0.2 \end{bmatrix}$$

$$\text{on 2nd epoch } \text{net}_1 = 0$$

$$\therefore W_2 = W_1 + \Delta W_1$$

$$= \begin{bmatrix} 1 & 2.0 & -0.2 \\ -1 & 1.0 & 0.4 \\ 0 & 0 & 0 \\ 0.5 & 0 & 0.2 \end{bmatrix}$$

$$\begin{bmatrix} 0.8 \\ -0.6 \\ 0 \\ 0.7 \end{bmatrix}$$

Step 2: Take second training pair  
 $X = X_2, d = d_2$

(1st epoch) : (1st) pair = 0.1

$$\therefore \text{net}_2 = W_2^t X$$

$$= \begin{bmatrix} 0.8 & -0.6 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix} = -1.6$$

$$O_2 = \text{sign}(-1.6)$$

$$= -1$$

$$\text{Here, } d = -1, O_2 = -1$$

$$\therefore (d - O_2) = 0$$

Hence,  $\Delta W_2 = 0$ , Thus there is no change in weights.

$$\therefore W_3 = W_2 = \begin{bmatrix} 0.8 \\ -0.6 \\ 0 \\ 0.7 \end{bmatrix}$$

Step 3: Take third training pair,

$$X = X_3, d = d_3$$

$$\text{net}_3 = W_3^t X$$

$$= \begin{bmatrix} 0.8 & -0.6 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0.5 \\ 1 \end{bmatrix} = -2.1$$

$$\therefore O_3 = \text{sign}(\text{net}_3) = \text{sign}(-2.1)$$

$$= -1$$

$$\therefore \Delta W_3 = c(d - 0.3) \times$$

$$= 0.1(1 - (-1))$$

$$\begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.2 \\ 0.2 \\ 0.1 \\ -0.2 \end{bmatrix}$$

$$W_4 = W_3 + \Delta W_3$$

$$= \begin{bmatrix} +0.8 \\ -0.6 \\ 0 \\ 0.7 \end{bmatrix} + \begin{bmatrix} -0.2 \\ 0.2 \\ 0.1 \\ -0.2 \end{bmatrix}$$

$$= \begin{bmatrix} 0.6 \\ -0.4 \\ 0.1 \\ 0.5 \end{bmatrix}$$

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Q. 12

$$X_1 = \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix}$$

$$X_2 = \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix}$$

$$X_3 = \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix}$$

$$d_1 = -1, d_2 = -1, d_3 = 1, c = 0.1, \lambda = 1$$

$$W_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0.5 \end{bmatrix}$$

Use delta learning rule to calculate the final weights. Use bipolar continuous activation function.

⇒ Sol<sup>n</sup>:-

For delta training with bipolar continuous activation function.

$$net_i = w_i^T x$$

$$o_i = F(net_i) = \frac{2}{1 + \exp(-\lambda net_i)} - 1$$

$$F'(net_i) = \frac{1}{2} (1 - o_i^2)$$

$$\Delta w_i = c (d - o_i) F'(net_i) x$$

Step 1: Take 1<sup>st</sup> training pair,  
 $X = x_1, d = d_1$

$$\text{net}_1 = W_1^+ X$$

$$= \begin{bmatrix} 1 & -1 & 0 & 0.5 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix} = 2.5$$

$$O = f(\text{net}_1)$$

$$= \frac{2}{1 + \exp(-1 \times 2.5)}$$

$$= \frac{2}{1 + 0.0820}$$

$$= 0.848$$

$$f'(\text{net}_1) = \frac{1}{2} (1 - O^2)$$

$$= \frac{1}{2} (1 - 0.848^2)$$

$$= 0.140$$

$$\Delta W_1 = c(d - O) f'(\text{net}_1) X$$

$$= 0.1(-1 - 0.848) \times 0.14 \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix}$$

$$= -0.026 \begin{bmatrix} 1 \\ -2 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} -0.026 \\ 0.052 \\ 0 \\ 0.026 \end{bmatrix}$$

$$W_2 = W_1 + \Delta W_1$$

$$= \begin{bmatrix} 1 \\ -1 \\ 0 \\ 0.5 \end{bmatrix} + \begin{bmatrix} -0.026 \\ 0.052 \\ 0 \\ 0.026 \end{bmatrix}$$

$$\therefore W_2 = \begin{bmatrix} 0.974 \\ -0.948 \\ 0 \\ 0.526 \end{bmatrix}$$

Step 2: Take 2<sup>nd</sup> pair,  $X = X_2$ ,  $d = d_2$

$$\begin{aligned} net_2 &= W_2^T X \\ &= \begin{bmatrix} 0.974 & -0.948 & 0 & 0.526 \end{bmatrix} \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix} = \end{aligned}$$

$$= (-1.948)$$

$$O_2 = F(net_2)$$

$$= \frac{2.590}{1 + \exp(-\lambda net_2)}$$

$$= \frac{1 + \exp(1.948)}{2} - 1$$

$$= \frac{1 + 7.0146}{2} - 1$$

$$= \frac{8.0146}{2} - 1$$

$$= 4.0073 - 1$$

$$= 3.0073$$

$$O_2 = 3.0073$$

$$\begin{aligned}
 f'(net_2) &= \frac{1}{2} (1 - o_2^2) \\
 &= \frac{1}{2} (1 - (-0.75)^2) \\
 &= 0.21875
 \end{aligned}$$

$$\Delta W_2 = c(d - o_2) f'(net_2) \times$$

$$= 0.1(-1 + 0.75)(0.21875) \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix}$$

$$= -0.0055 \begin{bmatrix} 0 \\ 1.5 \\ -0.5 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -0.0082 \\ 0.0028 \\ 0.0055 \end{bmatrix}$$

$$W_3 = W_2 + \Delta W_2$$

$$= \begin{bmatrix} 0.974 \\ -0.948 \\ 0 \\ 0.526 \end{bmatrix} + \begin{bmatrix} 0 \\ -0.0082 \\ 0.0028 \\ 0.0055 \end{bmatrix}$$

$$= \begin{bmatrix} 0.974 \\ -0.9562 \\ 0.0028 \\ 0.5315 \end{bmatrix}$$

Step 3: Take 3<sup>rd</sup> training pair;  
 $X = X_3$ ,  $d = d_3$

$$\text{net}_3 = W_3^T X = \begin{bmatrix} 0.974 & -0.9562 & 0.0028 & 0.5315 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0.5 \\ 1 \end{bmatrix}$$

$$= -2.4603$$

$$o_3 = F(\text{net}_3)$$

$$= \frac{2}{1 + \exp(2.4603)} - 1$$

$$= -0.8426$$

$$F'(\text{net}_3) = \frac{1}{2} (1 - o_3^2)$$

$$= \frac{1}{2} (1 - (-0.8426)^2)$$

$$= 0.145$$

$$\Delta W_3 = (0.1) (d - o_3) F'(\text{net}_3) \times$$

$$= (0.1) (1 - (-0.8426)) \times 0.145 \begin{bmatrix} -1 \\ 1 \\ 0.5 \\ -1 \end{bmatrix}$$

$$\Delta W_3 = 0.0267 \begin{bmatrix} -1 \\ -1 \\ 0.5 \\ -1 \end{bmatrix}$$

$$= \begin{bmatrix} -0.0267 \\ 0.0267 \\ 0.0133 \\ -0.0267 \end{bmatrix}$$

$$W_4 = W_3 + \Delta W_3$$

$$= \begin{bmatrix} 0.974 \\ -0.9562 \\ 0.0028 \\ 0.5315 \end{bmatrix} + \begin{bmatrix} -0.0267 \\ 0.0267 \\ 0.0133 \\ -0.0267 \end{bmatrix}$$

$$= \begin{bmatrix} 0.9473 \\ -0.9295 \\ 0.0161 \\ 0.504 \end{bmatrix}$$